

I build analyzers and reusable systems components in C++/C. Evidence includes the PS-form Analyzer, 12 AdvancedAlgorithms modules, an MCST internship, NASM IA-32 and a separate x86-64 codegen/register-allocation laboratory.

1st of 49

PS-form Analyzer, 104/104

12 C++23 modules

Graphs, trees, strings and data structures

Strict toolchain

CMake, -Werror, ASan/UBSan, clang-tidy

EXPERIENCE

JUL - AUG 2026	MCST - compiler engineering intern - Shared directed-graph core and RPO/cycle detection, Dijkstra, Tarjan SCC and Dinic components. - C++23/CMake, warnings-as-errors, ASan/UBSan, clang-tidy, explicit includes and I/O tests.
2024 - PRESENT	UniversalToolchain / Wist2 - creator and primary developer - Multiple IRs, interpreter/CIL execution paths and an opt-in AIR → SSA → AIR route. - Capability contracts, structural verifiers and parity tests; latest full run: 1,358/1,358 tests.

SELECTED PROJECTS

PS-form Memory Dependence Analyzer	C17 / PROGRAM ANALYSIS
Conservative yes/no/maybe semantics; normalization, range/residue/GCD filters, exact affine analysis, bounded search and exact/randomized/metamorphic verification.	
AdvancedAlgorithms	C++23 / HEADER-ONLY
12 reusable modules covering graphs, trees, strings and data structures; differential tests, structural invariants, sanitizers and large inputs.	
NASM IA-32 and x86-64 codegen	NASM COURSE
cdecl, stack frames, libc, structures/addressing and x87 in NASM IA-32; separate x86-64 SysV emitter with liveness, register allocation, iced-x86 and isolated native execution.	

CORE SKILLS

C++ / C
C++23, C17, reusable components, explicit contracts and error models
Algorithms
graphs, trees, strings, data structures, max-flow, SCC and shortest paths
Program analysis
conservative semantics, affine reasoning, exact oracles and counterexamples
Low-level
NASM IA-32, cdecl, stack frames and x87; x86-64 SysV codegen and disassembly
Toolchain
Linux, CMake, GitHub Actions, ASan/UBSan, clang-tidy and reproducible tests

EDUCATION AND AVAILABILITY

HSE University student. Moscow / remote. From September 2026: up to 20 hours per week.

TEACHING

Taught about 50 students in C/C++, Python and algorithms; authored a practical NASM IA-32 course.

RECOGNITION

HSE Olympiad prize-winner in competitive and industrial programming; top total score in MEPhI Junior (2025 Engineering Sciences; 2026 Information Technology); Baltic competition first-degree diploma and Grand Prize.